VYSHNAVI PENDELA

IoT DEVELOPER | EMBEDDED SYSTEMS

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PROFESSIONAL SUMMARY

Computer Science undergraduate graduating in 2027 with practical experimentation in ESP32-based prototypes, sensor interfacing and connected assistive / automation concepts. Familiar with Arduino IDE, GPIO-oriented hardware integration, serial debugging and alert logic using distance, time-of-flight and line sensors. Interested in IoT Developer, Embedded Systems Intern and Hardware-Software Integration roles.

TECHNICAL SKILLS

Boards & development: ESP32 Dev Module, ESP32-WROOM-32, Arduino IDE, embedded C / Arduino programming fundamentals
Sensors & modules: HC-SR04 ultrasonic, VL53L0X time-of-flight, TCRT-5000 IR line sensor, KY-038 sound sensor, Neo-6M GPS, R307 fingerprint sensor
Outputs & electronics: Buzzer, vibration motor, relay modules, L9110S / DRV8833 motor drivers, LCD interfacing
Engineering practice: Sensor interfacing, wiring and pin mapping, threshold-based alert logic, serial monitoring, hardware troubleshooting, prototype iteration
Supporting software: Python, basic SQL, Git / GitHub, Streamlit, web application fundamentals

SELECTED PROJECTS

- **AI Blind Assistance System | ESP32, sensors, alerts —** Assistive mobility prototype exploring obstacle and edge / fall-risk detection using ESP32, HC-SR04, VL53L0X and TCRT-5000 sensors. Worked on priority-based alert logic, including stop / warning behavior and buzzer or vibration feedback. Prototype integration and hardware documentation remain ongoing.
- **Smart Irrigation System | Python, Machine Learning —** AICTE-related project repository containing a Python application, irrigation dataset and trained model artefact for exploring data-informed irrigation decisions. (GitHub: Vyshnavi12548/Smart_Irrigation)
- **Biometric Attendance System | ESP32, R307 fingerprint module, web workflow —** Worked on a biometric attendance solution connecting fingerprint hardware with an attendance website. Project work included sensor communication, wiring and debugging the hardware-to-application workflow; final public repository link is to be added.
- **CloudCrop - Smart Rain Harvesting | IoT Product Concept —** Concept exploration for an IoT-enabled rainwater harvesting / monitoring solution for small businesses. Concept-stage; not presented as a completed deployment.

EXPERIENCE & PROGRAMMES

- **Edunet Foundation | AI / Data Analytics Internship Programme —** Remote, project-based learning that included work on data and an irrigation-related project repository.
- **ServiceNow | SmartBridge Virtual Internship —** Virtual internship learning programme focused on ServiceNow platform fundamentals.

EDUCATION

B.Tech, Computer Science and Engineering | TKR College of Engineering and Technology | Expected 2027 | CGPA: 8.16/10

SELECTED CERTIFICATIONS & LEARNING

Oracle - AI Foundations Associate | Oracle Cloud Infrastructure - Data Science Professional | IBM SkillsBuild - Artificial Intelligence Fundamentals with Capstone | IBM - Python for Data Science | Google Cloud Skill Badges - Prepare Data for ML APIs; Prompt Design in Vertex AI

Project and programme descriptions reflect documented repositories and previously shared experience.